KCE PATHWAYS · GRADE 3 · UNIT 5: DATA AND DESIGN · WEEK 23 · MATH

The Tile Workshop in Santa Fe — Teacher Lesson Plan & Answer Key

Learning objectives: Students identify quadrilaterals by their attributes (4 straight sides and 4 angles, closed), sort rectangles (4 right angles), rhombuses (4 equal sides) and squares (both), explain that a shape can belong to more than one group and that some quadrilaterals fit none of these groups, name equal parts of a shape's area with unit fractions, and solve two-step shape problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce Ms. Fresquez's adobe tile workshop in Santa Fe, New Mexico, with Emilio, Estrella and Felipa and a big tile order for a new courtyard.
2. Warm-Up: Measuring and Line Plots Review (10 pts, Week 22 skill): reading a ruler to the quarter inch and reading a line plot. If a student misses both, review "each small mark is one quarter inch; each X is one thing measured."
3. Picture It: The Tile Workshop in Santa Fe. The helpers count sides and angles, sort tiles into groups and cut tiles into equal parts. Ask: what do you notice about the corners and sides of these tiles?
4. Word Wrangler: 6 vocab cards (quadrilateral, attribute, right angle, rhombus, rectangle, unit fraction).
5. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W23.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
6. Meet the Quadrilaterals (10 pts): counting the quadrilaterals in a set of tiles, a find-the-mistake item (a curved side does not count) and a two-step sides problem. Tip: count the straight sides and check that the shape is closed.
7. Rectangles, Rhombuses, and Squares (10 pts): naming tiles by attributes. Tip: check two things, "Are all 4 angles right angles?" and "Are all 4 sides the same length?" Use the corner of a sheet of paper to test for right angles.
8. Shapes in More Than One Group (10 pts): the two sorting circles (4 right angles, 4 equal sides); squares go in the overlap, and a quadrilateral can fit neither circle.
9. Equal Parts and Unit Fractions (10 pts): naming the shaded unit fraction and checking that the parts have equal area. Equal parts of the same whole cover the same area even when their shapes differ.
10. Tile Challenge: Two-Step Problems (10 pts): counting right angles or sides across tiles, ribbon or trim around a shape, and the fraction not painted. Ask the student to name step 1 before solving.
11. Bonus Practice: The Drying Rack (XP only, not scored).
12. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches shape names to attributes, Set 2 matches equal parts to unit fractions, Set 3 matches shapes to how many sides or right angles they have.
13. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters for names of people and places, end mark, list commas, and one verb fix); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.

14. Speed Arena and Keyboard Corral (XP only, not scored).
15. Boss Battle, Part 1: The Courtyard Order (10 pts, 5 each): the most exact name for a tile, whether a rectangle is a rhombus, counting quadrilaterals, and a find-the-mistake item about equal sides and right angles.
16. Boss Battle, Part 2: The Last Tiles (10 pts, 5 each): naming a shaded unit fraction, two-step angle counts, and find-the-mistake items (count the parts, not the cuts; turning a square does not change it).
17. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W23.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
18. Results (Tile Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Tile Hero!” with the congratulations message “Tile hero! You sorted and split every tile in the order!” and a big confetti burst; below 85 shows “Great Effort Today!” with encouragement (“Every tile maker gets better with practice.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: MEASURING AND LINE PLOTS REVIEW (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Felipa measures a wooden strip for a tile frame. How long is the strip, to the nearest quarter inch? Type it like 2 1/2.	**3 3/4** (also accepted, with or without the unit word: 3 and 3/4, 3-3/4, 3 three fourths, 3 and three fourths, 3 and 3 fourths, 3.75, and 20 other spellings of the same answer)	The strip ends 3 quarter-inch marks past the 3. That is 3 3/4 inches.
w2_s0	Set 1: Which explanation is correct? The end of a paintbrush lines up with the mark exactly halfway between 5 and 6 on a ruler. [It is 5 1/2 inches long, because the mark halfway between 5 and 6 is 5 and one half. / It is 6 1/2 inches long, because the end is close to the 6. / It is 5 1/4 inches long, because every mark is a quarter inch. / It is 6 inches long, because we always round up.]	**It is 5 1/2 inches long, because the mark halfway between 5 and 6 is 5 and one half.**	Halfway between 5 and 6 is 5 1/2. The paintbrush has gone past 5 but has not reached 6.
w1_s1	Set 2: Felipa measured some broken tile chips and made this line plot. How many chips are longer than 1 1/4 inches?	**7** (also accepted: 7 chips, 7 tile chips)	Count the X marks to the right of 1 1/4: 4 at 1 1/2, 2 at 1 3/4, and 1 at 2. 4 + 2 + 1 = 7 chips.
w2_s1	Set 2: Find the mistake. The end of a nail is 1 small quarter-inch mark past the 2 on a ruler. Emilio says the nail is 2 3/4 inches long. What went wrong? [One quarter-inch mark past 2 is 2 1/4 inches, not 2 3/4. / Nothing is wrong. One mark past 2 is 2 3/4. / The nail is 3 inches long, because you count the next whole number. / The nail is 2 1/2 inches, because every mark is a half inch.]	**One quarter-inch mark past 2 is 2 1/4 inches, not 2 3/4.**	Each small mark is one quarter inch. One mark past 2 is 2 and 1 fourth inches.

ID	Question	Answer	Why (explain to student)
w1_s2	Set 3: Estrella measures a strip of trim for a tile. How long is it, to the nearest quarter inch? Type it like 2 1/2.	**1 1/4** (also accepted, with or without the unit word: 1 and 1/4, 1-1/4, 1 a fourth, 1 and a fourth, 1 and 1 fourth, 1.25, and 20 other spellings of the same answer)	The strip ends 1 quarter-inch mark past the 1. That is 1 1/4 inches.
w2_s2	Set 3: Which explanation is correct? On a line plot of tile lengths, there are 3 X marks above 2 1/2. What do they tell you? [3 tiles were each measured as 2 1/2 inches long. / One tile is 3 inches long. / The tiles are 2 1/2 inches long in all, added together. / 3 tiles are longer than 2 1/2 inches.]	**3 tiles were each measured as 2 1/2 inches long.**	On a line plot, each X stands for one measured object. 3 X marks above 2 1/2 means 3 tiles measured 2 1/2 inches.

MEET THE QUADRILATERALS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: How many of these tiles are quadrilaterals?	**3** (also accepted: 3 tiles, 3 quadrilaterals)	Tiles A, C, and E each have 4 straight sides. Tile B has 3 sides and tile D has 5 sides. That is 3 quadrilaterals.
t2_s0	Set 1: Find the mistake. Felipa finds a tile with 3 straight sides and 1 curved side. She says it is a quadrilateral because it has 4 sides. What went wrong? [A quadrilateral needs 4 straight sides. A curved side does not count, so this tile is not a quadrilateral. / Nothing is wrong. Any shape with 4 sides is a quadrilateral. / It is a triangle, because it has 3 straight sides. / It is a rectangle, because it has 4 sides.]	**A quadrilateral needs 4 straight sides. A curved side does not count, so this tile is not a quadrilateral.**	Every side of a quadrilateral must be a straight line. A shape with a curved side is not a quadrilateral.
t1_s1	Set 2: How many of these tiles are quadrilaterals?	**2** (also accepted: 2 tiles, 2 quadrilaterals)	Tiles A and C each have 4 straight sides. B has 6 sides, D has 3 sides, and E has 5 sides. That is 2 quadrilaterals.
t2_s1	Set 2: Two steps! Ms. Fresquez puts 3 triangle tiles and 5 quadrilateral tiles on a tray. How many sides do all the tiles have in all?	**29** (also accepted: 29 sides)	Step 1: 3 triangles × 3 sides = 9 sides, and 5 quadrilaterals × 4 sides = 20 sides. Step 2: 9 + 20 = 29 sides.
t1_s2	Set 3: How many of these tiles are quadrilaterals?	**3** (also accepted: 3 tiles, 3 quadrilaterals)	Tiles A, B, and E each have 4 straight sides. C has 3 sides and D has 6 sides. That is 3 quadrilaterals.
t2_s2	Set 3: Which explanation is correct? Why is a triangle not a quadrilateral? [A triangle has only 3 sides and 3 angles. A quadrilateral must have 4 sides and 4 angles. / A triangle is too small to be a quadrilateral. / A triangle has no angles at all. / A triangle has curved sides.]	**A triangle has only 3 sides and 3 angles. A quadrilateral must have 4 sides and 4 angles.**	Quadrilaterals have exactly 4 straight sides and 4 angles. A triangle has 3 of each.

RECTANGLES, RHOMBUSES, AND SQUARES (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: What is the best name for this tile: rectangle, rhombus, or square? The tick marks show sides that are the same length.	**rhombus** (also accepted: a rhombus, an rhombus, it is a rhombus, it's a rhombus, it is an rhombus)	All 4 sides are the same length, so it is a rhombus. The corners are not square corners, so it is not a rectangle or a square.
r2_s0	Set 1: A tile has 4 right angles. Its sides are 6 inches, 3 inches, 6 inches, and 3 inches long. Is it a rectangle, a rhombus, or a square?	**rectangle** (also accepted: a rectangle, an rectangle, it is a rectangle, it's a rectangle, it is an rectangle)	It has 4 right angles, so it is a rectangle. Its sides are not all the same length, so it is not a rhombus or a square.
r1_s1	Set 2: This tile has 4 right angles and 4 equal sides. Type the most exact name for it.	**square** (also accepted: a square, an square, it is a square, it's a square, it is an square)	A shape with 4 right angles and 4 equal sides is a square. (It is also a rectangle and a rhombus, but square is the most exact name.)
r2_s1	Set 2: Find the mistake. Emilio says, "A square cannot be a rectangle, because a rectangle has 2 long sides and 2 short sides." What went wrong? [A rectangle only needs 4 right angles. A square has 4 right angles, so a square is a special kind of rectangle. / Nothing is wrong. Rectangles must have 2 long sides and 2 short sides. / A square is a triangle, not a rectangle. / A square has no right angles, so it cannot be a rectangle.]	**A rectangle only needs 4 right angles. A square has 4 right angles, so a square is a special kind of rectangle.**	The rule for a rectangle is 4 right angles. Side lengths can be different or all the same, so a square counts.
r1_s2	Set 3: What is the best name for this tile: rectangle, rhombus, or square?	**rectangle** (also accepted: a rectangle, an rectangle, it is a rectangle, it's a rectangle, it is an rectangle)	It has 4 right angles, so it is a rectangle. Its long sides and short sides are different lengths, so it is not a square or a rhombus.
r2_s2	Set 3: Which explanation is correct? Is every rhombus a square? [No. A rhombus has 4 equal sides, but it is only a square if it also has 4 right angles. / Yes. Every shape with 4 sides is a square. / Yes. A rhombus and a square are always the same shape. / No. A rhombus has 3 sides.]	**No. A rhombus has 4 equal sides, but it is only a square if it also has 4 right angles.**	Some rhombuses have square corners and some do not. Only the ones with 4 right angles are squares.

SHAPES IN MORE THAN ONE GROUP (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
g1_s0	Set 1: How many of these names fit a square: quadrilateral, rectangle, rhombus, triangle?	**3** (also accepted: 3 names)	A square has 4 sides (quadrilateral), 4 right angles (rectangle), and 4 equal sides (rhombus). It is not a triangle. That is 3 names.
g2_s0	Set 1: Look at the sorting circles. Which tile belongs outside both circles? [A tile with sides of 3, 5, 4, and 6 inches and no right angles / A tile with sides of 2, 7, 2, and 7 inches and 4 right angles / A tile with 4 sides of 5 inches each and no right angles / A tile with 4 sides of 5 inches each and 4 right angles]	**A tile with sides of 3, 5, 4, and 6 inches and no right angles**	The tile with sides 3, 5, 4, and 6 inches has no right angles and no equal sides, so it fits neither circle. It is still a quadrilateral.

ID	Question	Answer	Why (explain to student)
g1_s1	Set 2: Think carefully! Ms. Fresquez has 6 square tiles and 4 rhombus tiles that are not squares. How many of these tiles have 4 equal sides?	**10** (also accepted: 10 tiles)	Squares have 4 equal sides too. 6 squares + 4 other rhombuses = 10 tiles with 4 equal sides.
g2_s1	Set 2: Find the mistake. Estrella says, "This tile has 4 straight sides, but it is not a rectangle, a rhombus, or a square, so it is not a quadrilateral." What went wrong? [Any closed shape with 4 straight sides is a quadrilateral, even if it is not a rectangle, a rhombus, or a square. / Nothing is wrong. Only rectangles, rhombuses, and squares are quadrilaterals. / It is a triangle, because it is not a square. / It is a hexagon, because its sides are different.]	**Any closed shape with 4 straight sides is a quadrilateral, even if it is not a rectangle, a rhombus, or a square.**	Rectangles, rhombuses, and squares are just some of the quadrilaterals. Many quadrilaterals fit none of those groups.
g1_s2	Set 3: Think carefully! Ms. Fresquez has 5 rectangle tiles that are not squares and 3 square tiles. How many of these tiles have 4 right angles?	**8** (also accepted: 8 tiles)	Squares have 4 right angles too. 5 rectangles + 3 squares = 8 tiles with 4 right angles.
g2_s2	Set 3: Which statement is true? [Every square is a rhombus. / Every rhombus is a square. / Every rectangle is a square. / Every quadrilateral is a rectangle.]	**Every square is a rhombus.**	A square always has 4 equal sides, so it is always a rhombus. The other statements are only true some of the time.

EQUAL PARTS AND UNIT FRACTIONS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: Emilio split this tile into equal parts. What fraction of the tile's area is shaded?	**1/6** (also accepted: 1 / 6, 1 sixth, one sixth, one-sixth, a sixth)	The tile is split into 6 equal parts and 1 part is shaded, so the shaded part is 1/6 of the area.
e2_s0	Set 1: Find the mistake. Emilio cut this tile into 4 parts. He says each part is 1/4 of the tile. What went wrong? [The 4 parts are not the same size, so they are not fourths. Fourths must have equal areas. / Nothing is wrong. Any 4 parts are fourths. / Each part is 1/3, because there are 3 lines. / Each part is 1/2, because the tile is a rectangle.]	**The 4 parts are not the same size, so they are not fourths. Fourths must have equal areas.**	A fraction like 1/4 only works when all the parts have the same area. These parts are different sizes.
e1_s1	Set 2: What fraction of this square tile's area is shaded?	**1/8** (also accepted: 1 / 8, 1 eighth, one eighth, one-eighth, a eighth)	The square is split into 8 equal triangles and 1 is shaded, so the shaded part is 1/8 of the area.
e2_s1	Set 2: Two steps! Estrella splits a square tile into 4 equal parts. Then she splits each part in half. What fraction of the whole tile is one small part?	**1/8** (also accepted: 1 / 8, 1 eighth, one eighth, one-eighth, a eighth)	Step 1: 4 equal parts. Step 2: each part split in half makes 4 × 2 = 8 equal parts. Each small part is 1/8 of the tile.
e1_s2	Set 3: What fraction of this tile's area is shaded?	**1/3** (also accepted: 1 / 3, 1 third, one third, one-third, a third)	The tile is split into 3 equal parts and 1 part is shaded, so the shaded part is 1/3 of the area.

ID	Question	Answer	Why (explain to student)
e2_s2	Set 3: Which explanation is correct? One square tile is cut into 4 equal triangles. A same-size square tile is cut into 4 equal strips. Does one triangle cover the same area as one strip? [Yes. Each one is 1/4 of the same-size square, so they cover the same area even though their shapes are different. / No. Triangles are always smaller than strips. / No. The strip is bigger because it is longer. / Yes, but only because both tiles are blue.]	**Yes. Each one is 1/4 of the same-size square, so they cover the same area even though their shapes are different.**	Equal parts of the same whole have the same area. Each piece is 1/4 of a same-size square, so the areas match.

TILE CHALLENGE: TWO-STEP PROBLEMS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: Two steps! Square tiles and rectangle tiles each have 4 right angles. Felipa packs 3 square tiles and 4 rectangle tiles in a box. How many right angles are in the box in all?	**28** (also accepted: 28 right angles, 28 angles)	Step 1: 3 + 4 = 7 tiles. Step 2: 7 tiles × 4 right angles = 28 right angles.
c2_s0	Set 1: Two steps! A rhombus tile has one side that is 3 inches long. Ms. Fresquez glues ribbon along all 4 edges. How many inches of ribbon does she need?	**12** (also accepted: 12 inches, 12 in)	Step 1: A rhombus has 4 equal sides, so every side is 3 inches. Step 2: 3 + 3 + 3 + 3 = 12 inches.
c1_s1	Set 2: Two steps! Emilio splits a square tile into 8 equal parts. He paints 3 parts red. What fraction of the tile is not painted?	**5/8** (also accepted: 5 / 8, 5 eighths, five eighths, five-eighths)	Step 1: 8 − 3 = 5 parts are not painted. Step 2: each part is 1/8, so 5 parts is 5/8 of the tile.
c2_s1	Set 2: Two steps! A rectangle tile is 6 inches long and 2 inches wide. Felipa adds the lengths of all 4 sides to know how much trim to buy. How many inches of trim?	**16** (also accepted: 16 inches, 16 in)	Step 1: In a rectangle, sides across from each other match: 6, 2, 6, and 2 inches. Step 2: 6 + 2 + 6 + 2 = 16 inches.
c1_s2	Set 3: Two steps! Ms. Fresquez has 4 rhombus tiles and 2 triangle tiles. How many sides do the tiles have in all?	**22** (also accepted: 22 sides)	Step 1: 4 rhombuses × 4 sides = 16 sides, and 2 triangles × 3 sides = 6 sides. Step 2: 16 + 6 = 22 sides.
c2_s2	Set 3: Two steps! A tray holds 3 rows of tiles with 5 tiles in each row. 4 of the tiles are rhombuses and the rest are squares. How many tiles are squares?	**11** (also accepted: 11 squares, 11 tiles, 11 square tiles)	Step 1: 3 rows × 5 tiles = 15 tiles. Step 2: 15 − 4 = 11 squares.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match quadrilateral	**a flat shape with 4 straight sides**	2 points if matched on the first try.
match_s0_2	Set 1: Match rhombus	**4 sides that are all the same length**	2 points if matched on the first try.
match_s0_3	Set 1: Match rectangle	**4 right angles**	2 points if matched on the first try.
match_s0_4	Set 1: Match square	**4 equal sides and 4 right angles**	2 points if matched on the first try.
match_s0_5	Set 1: Match right angle	**a square corner**	2 points if matched on the first try.
match_s1_1	Set 2: Match 2 equal parts	**each part is 1/2**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s1_2	Set 2: Match 3 equal parts	**each part is 1/3**	2 points if matched on the first try.
match_s1_3	Set 2: Match 4 equal parts	**each part is 1/4**	2 points if matched on the first try.
match_s1_4	Set 2: Match 6 equal parts	**each part is 1/6**	2 points if matched on the first try.
match_s1_5	Set 2: Match 8 equal parts	**each part is 1/8**	2 points if matched on the first try.
match_s2_1	Set 3: Match Sides on a triangle	**3**	2 points if matched on the first try.
match_s2_2	Set 3: Match Right angles on a square	**4**	2 points if matched on the first try.
match_s2_3	Set 3: Match Sides on a hexagon	**6**	2 points if matched on the first try.
match_s2_4	Set 3: Match Sides on a pentagon	**5**	2 points if matched on the first try.
match_s2_5	Set 3: Match Right angles on a rhombus with no square corners	**0**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: the tiles came from santa fe	**The tiles came from Santa Fe.**	Start with a capital T. Santa Fe is the name of a city, so both words get capital letters. End the telling sentence with a period.
fs2_s0	Set 1: Fix the sentence: we cut squares rectangles and rhombuses	**We cut squares, rectangles, and rhombuses.** (also accepted: We cut squares, rectangles and rhombuses.)	Start with a capital W, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: is every square a rectangle	**Is every square a rectangle?**	Start with a capital I. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: emilio estrella and felipa painted the tiles	**Emilio, Estrella, and Felipa painted the tiles.** (also accepted: Emilio, Estrella and Felipa painted the tiles.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: how many right angles does a square have	**How many right angles does a square have?**	Start with a capital H. The sentence asks something, so it ends with a question mark.
fs2_s2	Set 3: Fix the sentence: a rhombus have four equal sides	**A rhombus has four equal sides.**	Start with a capital A and end with a period. One rhombus means use has, not have.

BOSS BATTLE: THE COURTYARD ORDER (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Ms. Fresquez's newest tile has 4 right angles and 4 sides that are each 7 inches long. Type the most exact name for it.	**square** (also accepted: a square, an square, it is a square, it's a square, it is an square)	It has 4 right angles and 4 equal sides, so the most exact name is square.

ID	Question	Answer	Why (explain to student)
bb2_s0	Set 1: Which explanation is correct? Felipa says a rectangle that is 8 inches long and 3 inches wide is also a rhombus. [Felipa is wrong. A rhombus needs 4 equal sides, and this rectangle's sides are 8, 3, 8, and 3 inches. / Felipa is right. Every rectangle is a rhombus. / Felipa is right, because it has 4 right angles. / Felipa is wrong, because a rectangle has 3 sides.]	**Felipa is wrong. A rhombus needs 4 equal sides, and this rectangle's sides are 8, 3, 8, and 3 inches.**	A rhombus must have 4 sides of the same length. 8 inches and 3 inches are different lengths.
bb1_s1	Set 2: Two steps! Estrella sorts 9 tiles. 4 are squares, 3 are rhombuses that are not squares, and the rest are triangles. How many of the tiles are quadrilaterals?	**7** (also accepted: 7 tiles, 7 quadrilaterals)	Squares and rhombuses both have 4 sides. 4 + 3 = 7 quadrilaterals. (The other 9 − 7 = 2 tiles are triangles.)
bb2_s1	Set 2: Find the mistake. Emilio says, "Any tile with 4 equal sides must have 4 right angles." What went wrong? [A rhombus can have 4 equal sides with no square corners at all, like a diamond shape. / Nothing is wrong. Equal sides always make right angles. / A tile with 4 equal sides is a triangle. / A tile with 4 equal sides has 8 right angles.]	**A rhombus can have 4 equal sides with no square corners at all, like a diamond shape.**	Equal sides make a rhombus. A rhombus only has right angles when it is also a square.
bb1_s2	Set 3: Felipa cuts a tile into equal parts. Each part is 1/6 of the tile's area. How many parts are there?	**6** (also accepted: 6 parts, 6 equal parts)	A unit fraction of 1/6 means the whole is split into 6 equal parts.
bb2_s2	Set 3: Which shape belongs in both groups: "4 right angles" and "4 equal sides"? [A square / A rhombus with no square corners / A rectangle that is 9 inches long and 2 inches wide / A triangle]	**A square**	Only a square has 4 right angles and 4 equal sides at the same time.

BOSS BATTLE: THE LAST TILES (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: What fraction of this square tile's area is the shaded triangle?	**1/4** (also accepted: 1 / 4, 1 fourth, one fourth, one-fourth, a fourth)	Lines from corner to corner split the square into 4 equal triangles. One is shaded, so it is 1/4 of the area.
bb4_s0	Set 1: Find the mistake. Emilio cuts a square tile into 8 equal parts and paints 1 part. He says 1/7 of the tile is painted, because 7 parts are left. Type the fraction that is painted.	**1/8** (also accepted: 1 / 8, 1 eighth, one eighth, one-eighth, a eighth)	The whole tile has 8 equal parts, so the bottom number is 8. 1 painted part is 1/8 of the tile.
bb3_s1	Set 2: Two steps! A tray has 4 square tiles and 3 rhombus tiles that have no square corners. How many right angles are on the tray?	**16** (also accepted: 16 right angles, 16 angles)	Step 1: 4 squares × 4 right angles = 16. Step 2: the 3 rhombuses have 0 right angles, so 16 + 0 = 16 right angles.
bb4_s1	Set 2: Find the mistake. Felipa makes 5 straight cuts across a long tile and gets 6 equal parts. She says each part is 1/5, because she made 5 cuts. Type the correct fraction for one part.	**1/6** (also accepted: 1 / 6, 1 sixth, one sixth, one-sixth, a sixth)	Count the parts, not the cuts. 5 cuts make 6 equal parts, so each part is 1/6.
bb3_s2	Set 3: Two steps! Ms. Fresquez makes 6 rectangle tiles and 2 triangle tiles. How many angles do the tiles have in all?	**30** (also accepted: 30 angles)	Step 1: 6 rectangles × 4 angles = 24, and 2 triangles × 3 angles = 6. Step 2: 24 + 6 = 30 angles.

ID	Question	Answer	Why (explain to student)
bb4_s2	Set 3: Find the mistake. Emilio says this tile is not a square because it is tipped on its corner. The marks show 4 equal sides and 4 right angles. Type the most exact name for the tile.	**square** (also accepted: a square, an square, it is a square, it's a square, it is an square)	Turning a shape does not change its attributes. It still has 4 equal sides and 4 right angles, so it is a square.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W23.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: strip ends 2 quarter-inch marks past the 3	3 2/4 inches, which is the same as 3 1/2 inches. Two quarter inches make a half inch.
2	Draw a tile shape with 4 straight sides	Any closed shape with 4 straight sides (a quadrilateral).
3	Sides on a triangle; sides on a rectangle	A triangle has 3 sides; a rectangle has 4 sides.
4	Which tile has 4 square corners?	a rectangle tile
5	Square in 2 equal parts, 1 shaded	1/2. Check that the two parts are the same size.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Draw a rhombus that is not a square; mark the equal sides	A tilted, diamond-like shape with 4 equal sides (each marked) and no right angles.
2	4 right angles, 4 sides each 5 inches: circle every name	quadrilateral, rectangle, rhombus and square. Not a triangle.
3	2 square tiles and 5 triangle tiles: angles in all	Step 1: 2 × 4 = 8 angles and 5 × 3 = 15 angles. Step 2: 8 + 15 = 23 angles.
4	Emilio’s 4 pieces of different sizes are each 1/4	The pieces are not equal, so they are not fourths; fourths must have equal areas. The drawing should show the rectangle cut into 4 equal parts.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Square tile in 8 equal parts, 1 shaded	1/8. Check that all 8 parts are the same size.